

Practical 9:

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
GNU nano 8.7
int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3], n=12, frames=3;
    int i,j,k,hit=0,fault=0,pos,far;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            for(j=0;j<frames;j++) {
                if(frame[j] == -1) {
                    frame[j] = pages[i];
                    found = 1;
                    break;
                }
            }

            if(!found) {
                far = -1;
                pos = -1;

                for(j=0;j<frames;j++) {
                    int next = -1;

                    for(k=i+1;k<n;k++) {
                        if(frame[j] == pages[k]) {
                            next = k;
                            break;
                        }
                    }

                    if(next == -1) {
                        pos = j;
                        break;
                    }

                    if(next > far) {
                        far = next;
                        pos = j;
                    }
                }

                frame[pos] = pages[i];
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n", (float)hit/n*100, (float)fau1t/n*100);

    return 0;
}
```

Output:

```
M ~  
Admin@DESKTOP-PNDN9B0 MSYS ~  
$ nano practical9_optimal.c  
Admin@DESKTOP-PNDN9B0 MSYS ~  
$ gcc practical9_optimal.c -o optimal  
./optimal  
Page      Frames  
1          1 - -  
2          1 2 -  
3          1 2 3  
4          1 2 4  
1          1 2 4  
2          1 2 4  
5          1 2 5  
1          1 2 5  
2          1 2 5  
3          3 2 5  
4          4 2 5  
5          4 2 5  
  
Total Hits = 5  
Total Faults = 7  
Hit Percentage = 41.67%  
Fault Percentage = 58.33%  
Admin@DESKTOP-PNDN9B0 MSYS ~
```